Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

Identify Z in the sequence

$$CH_3COONH_4 \xrightarrow{\Delta} X \xrightarrow[\Delta]{P_2O_5} Y \xrightarrow{H_2O/H^+} Z$$

(A) $CH_3CH_2CONH_2$

(B) CH_3CN

(C) CH_3COOH

(D) $(CH_3CO)_2O$

Q.2

A colourless liquid, at room temperature, reacts with soda-lime to form sodium salt of a carboxylic acid and ammonia gas. The liquid is

(A) propanoic acid

(B) formamide

(C) propanamide

(D) methyl enthanoate

Q.3

Which of the following on oxidation followed by hydrolysis gives pyruvic acid?

(A) Acetaldehyde cyanohydrin

(B) Acetone cyanohydrin

(C) Formaldehyde cyanohydrin

(D) None of these

Q.4

Identify the correct order of boiling points of the following compounds:

$CH_3CH_2CH_2CH_2OH$ $CH_3CH_2CH_2CHO$

1 2

$CH_3CH_2CH_2COOH$

3

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(A) $1 > 2 > 3$

(B) $3 > 1 > 2$

(C) $1 > 3 > 2$

(D) $3 > 2 > 1$

Q.5

Carboxylic acids are more acidic than phenol and alcohol because of

(A) intermolecular hydrogen bonding

(B) formation of dimers

(C) highly acidic hydrogen

(D) resonance stabilization of their conjugate base

Q.6

Which of the following compounds will react with $NaHCO_3$ solution to give sodium salt and carbon dioxide?

(A) Acetic acid

(B) n -Hexanol

(C) Phenol

(D) Both (b) and (c)

Q.7

Which of the following acids has the smallest dissociation constant?

(A) $CH_3CH(F)COOH$

(B) FCH_2CH_2COOH

(C) $BrCH_2CH_2COOH$

(D) $CH_3CH(Br)COOH$

Q.8

Which of the following orders of relative strengths of acids is correct?

(A) $FCH_2COOH > ClCH_2COOH > BrCH_2COOH$

(B) $ClCH_2COOH > BrCH_2COOH > FCH_2COOH$

(C) $BrCH_2COOH > ClCH_2COOH > FCH_2COOH$

(D) $ClCH_2CO_2H > FCH_2COOH > BrCH_2COOH$

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Q.9

An ester is boiled with KOH. The product is cooled and acidified with concentrated HCl. A white crystalline acid separates. The ester is

(A) methyl acetate

(B) ethyl acetate

(C) ethyl formate

(D) ethyl benzoate

Q.10

The correct product of the following sequence of reactions

$$(CH_3)_2CHCOOH \xrightarrow[(ii)\ H_2O]{(i)\ LiAlH_4} \xrightarrow{PBr_3} \xrightarrow[DMSO]{KCN} \xrightarrow[\Delta]{H_2O^+,\ H^+}$$

(A) $(CH_3)_2\ CHCHBrCOOH$

(B) $(CH_3)_2\ CHCH_2COOH$

(C) $(CH_3)_2\ CHCH_2CH_2NH_2$

(D) $(CH_3)_2\ C = CHCOOH$

Q.11

Which one of the following orders of acid strength is correct?

(A) $RCOOH > HOH > ROH > HC \equiv CH$

(B) $RCOOH > HOH > HC \equiv CH > ROH$

(C) $RCOOH > HC \equiv CH > HOH > ROH$

(D) $RCOOH > ROH > HOH > HC \equiv CH$

Q.12

$$(CH_3)_2CO \xrightarrow[(HCl)]{NaCN} A \xrightarrow[\Delta]{H_3O^+} B$$

In the above sequence of reactions A and B are

(A) $(CH_3)_2\ C(OH)CN, (CH_3)_2\ C(OH)COOH$

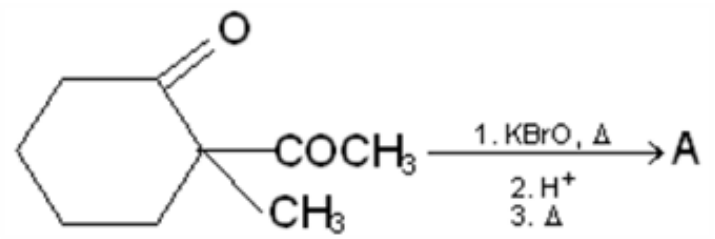
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(B) $(CH_3)_2C(OH)CN, (CH_3)_2C(OH)_2$

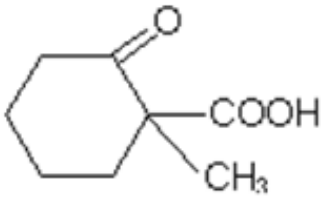
(C) $(CH_3)_2C(OH)CN, (CH_3)_2CHCOOH$

(D) $(CH_3)_2C(OH)CN, (CH_3)_2C=O$

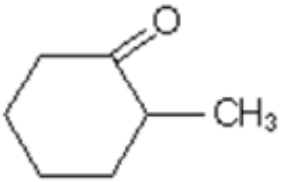
Q.13



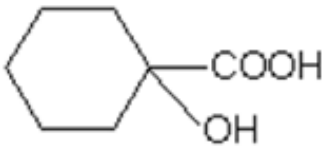
(A)



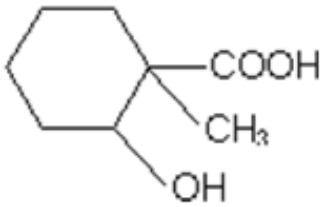
(B)



(C)



(D)



Q.14

The carboxyl functional group $(-COOH)$ is present in

(A) picric acid

(B) barbituric acid

(C) ascorbic acid

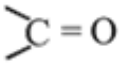
(D) aspirin

Q.15

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Carboxylic acid group does not give the usual addition and elimination reactions of aldehydes and ketones because

(A) O-H bond is more polar than



group

(B) carboxylate ion gets ionised

(C) carboxylate ion gets stabilised by resonance

(D) it exists as $-COOH$ and there is no carbonyl group

Q.16

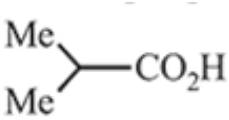
The correct order of increasing acid strength of the compounds

(A) CH_3CO_2H

(B) $MeOCH_2CO_2H$

(C) CF_3CO_2H

(D)



is

(A) $D < A < B < C$

(B) $A < D < B < C$

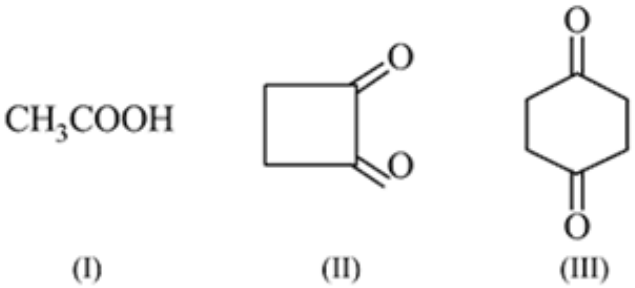
(C) $B < D < A < C$

(D) $D < A < C < B$

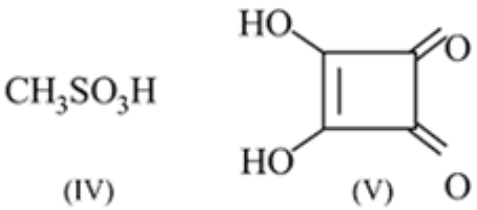
Q.17

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Number of compounds having hydrogen more acidic than cyclo hexan-1,3-dione



(I) (II) (III)



(IV) (V)

(A) 3

(B) 4

(C) 2

(D) 5

Q.18

Identify Z in the following sequence of reactions-

$$CH_3COONH_4 \xrightarrow{\Delta} X \xrightarrow{P_2O_5} Y \xrightarrow{H_2O/H} Z$$

(A) $CH_3 - CH_2 - CO - NH_2$

(B) $CH_3 - CN$

(C) $(CH_3CO)_2\,O$

(D) $CH_3 - COOH$

Q.19

Reaction between solid ice and methyl magnesium bromide gives an addition compound which or acidification yields:

(A) ethanoic acid

(B) propanoic acid

(C) ethanal

(D) ethanol.

Q.20

H-bonding is the reason for dimerisation of:

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(A) $NO_2(s)$

(B) $BeCl_2(s)$

(C) $CH_3COOH\ (g)$

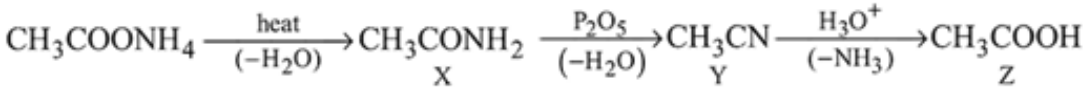
(D) $H_3BO_3(s)$

Answers & Solutions

Q.1 Answer:

CH_3COOH

Solution:



Q.2 Answer:

formamide

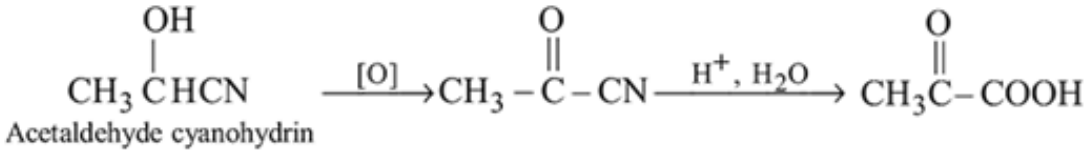
Solution:

Only amides (but not acids and esters) undergo hydrolysis in presence of soda-lime to form sodium salt of a carboxylic acid and ammonia gas. Further, since the given compound is a liquid, it must be formamide, because propanamide is a solid.

Q.3 Answer:

Acetaldehyde cyanohydrin

Solution:



Q.4 Answer:

$3 > 1 > 2$

Solution:

Due to $-I$ effect of the - COOH group, H-bonds in acids are much stronger than in alcohols; while aldehydes do not exhibit H -bonding.

Q.5 Answer:

resonance stabilization of their conjugate base

Solution:

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In carboxylates (conjugate base of carboxylic acids), resonance is more significant because the two resonating structures are similar, while in phenoxide, the resonating structures are not equivalent, alkoxide ions do not show resonance.

Q.6 Answer:

Acetic acid

Solution:

Only acetic acid is more acidic than carbonic acid (conjugate acid of $NaHCO_3$) hence it dissolves in $NaHCO_3$,

$$CH_3COOH + NaHCO_3 \longrightarrow CH_3COO^-Na^+ + H_2CO_3\ (H_2O + CO_2)$$

stronger acid **weaker acid**

while phenol and n -hexanol are less acidic than carbonic acid and hence do not dissolve in $NaHCO_3$

Q.7 Answer:

$BrCH_2CH_2COOH$

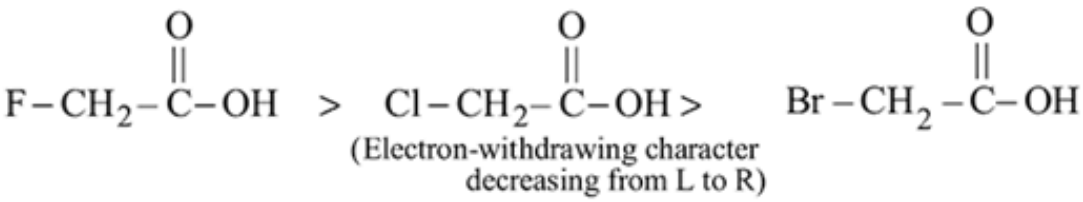
Solution:

Bromine is less electronegative than F, further in $BrCH_2CH_2COOH$, Br is more away from the $-COOH$ group than in $CH_3CHBrCOOH$.

Q.8 Answer:

$FCH_2COOH > ClCH_2COOH > BrCH_2COOH$

Solution:



Q.9 Answer:

ethyl benzoate

Solution:

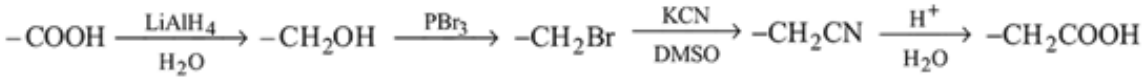
Ethyl benzoate on hydrolysis gives benzoic acid which is a solid, other esters give CH_3COOH and HCOOH, both of which are liquids.

Q.10 Answer:

$(CH_3)_2\ CHCH_2COOH$

Solution:

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Q.11 Answer:

$RCOOH > HOH > ROH > HC \equiv CH$

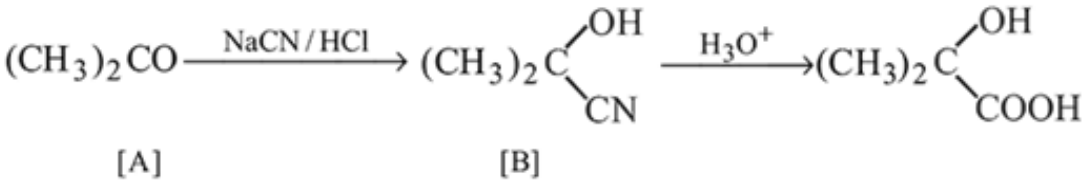
Solution:

$RCOOH > ROH > HC \equiv CH$. **Water is more acidic then alcohol.**

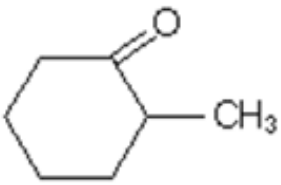
Q.12 Answer:

$(CH_3)_2 C(OH)CN, (CH_3)_2 C(OH)COOH$

Solution:

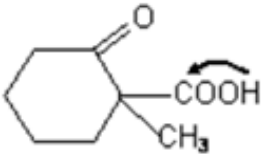


Q.13 Answer:

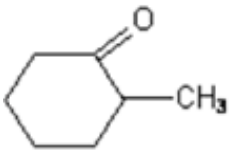


Solution:

Intermediate is



which decarboxylates to



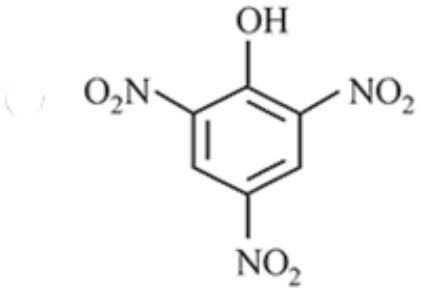
(when β -keto acid is heated, CO_2 is lost)

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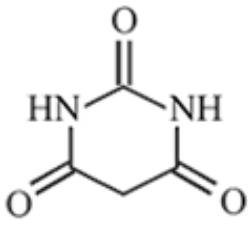
Q.14 Answer:

aspirin

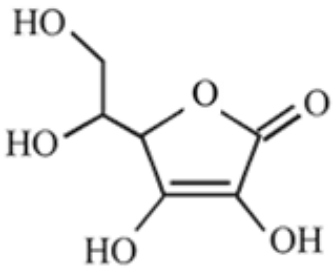
Solution:



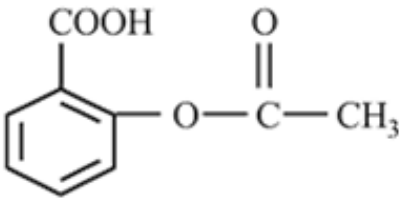
(Picric acid)



(Barbituric acid)



(Ascorbic acid)



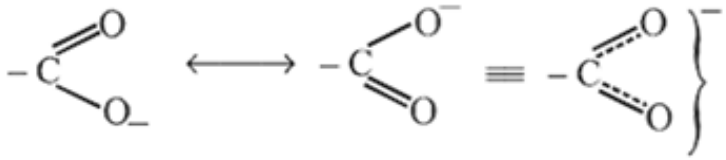
(Aspirin)

Q.15 Answer:

carboxylate ion gets stabilised by resonance

Solution:

Due to resonance in carboxylate ion, the double bond character of $C = O$ bond in carboxylic acids is greatly reduced as compared to that in aldehydes and ketones.



Q.16 Answer:

$D < A < B < C$

Solution:

The correct order of increasing acid strength $(Me)_2CH.COOH < CH_3COOH < MeOCH_2COOH < CF_3COOH$

Electron withdrawing groups increase the acid strength and electron donating groups decrease the acid strength.

Q.17 Answer:

3

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Solution:

Compounds I, IV and V will have more acidic hydrogen.

Q.18 Answer:

$CH_3 - COOH$

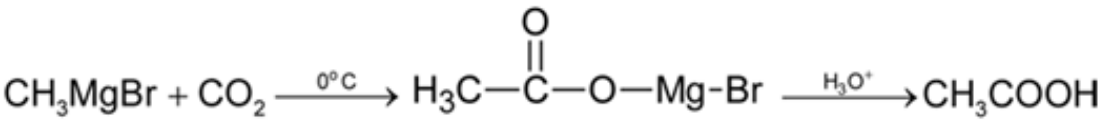
Solution:

$$CH_3COONH_4 \xrightarrow{\Delta} \underset{\textbf{(X)}}{CH_3CONH_2} \xrightarrow{\Delta/P_2O_5} \underset{\textbf{(Y)}}{CH_3CN} \xrightarrow{H_3O^{\oplus}} \underset{\textbf{(Z)}}{CH_3COOH}$$

Q.19 Answer:

ethanoic acid

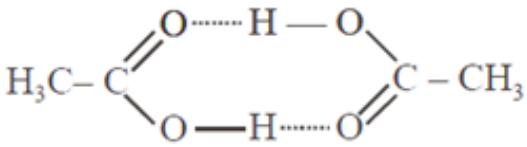
Solution:



Q.20 Answer:

$CH_3COOH \ (g)$

Solution:



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